

2	(a)	The acceleration of the particle is proportional to its displacement from the equilibrium position and is always directed towards that position.	B1 B1
	(b)	At $y = 0.000 \text{ m}$, $TE = KE + GPE + EPE$ $= 0 + 0 + 0.445$ $= 0.445 \text{ J}$ <i>-Accept answer between 0.440 J and 0.445 J.</i>	C1 A1
	(c)	Straight line passing through the origin and (0.393, 0.160). 1. At $y = 0.160 \text{ m}$, $TE = KE + GPE + EPE$ $0.445 = 0 + GPE + 0.049 \Rightarrow GPE = 0.396 \text{ J}$ <i>Acceptable value for the GPE at $y = 0.160 \text{ m}$ ranges from 0.390 J to 0.396 J.</i> 2. <i>Students who draw curves get zero mark.</i>	B1 B1
	(d)	<u>Method 1:</u> Consider (0.393, 0.160) on GPE line,	

	$mg (0.160) = 0.393$ $m = 0.250 \text{ kg}$ <u>Method 2:</u> At $y = 0.080 \text{ m}$, the equilibrium position of the SHM, extension e of spring is 0.040 m . (See Fig. 4.1). There is no net force on m , thus $ke = mg$ $(61.4)(0.040) = m (9.81) \Rightarrow m = 0.250 \text{ kg}$	C2 A0
(e)	<u>Method 1:</u> $KE_{\max} = \frac{1}{2}mv_o^2 = \frac{1}{2}m\omega^2x_o^2 \quad \Rightarrow \quad 0.196 = \frac{1}{2}(0.250)\left(\frac{2\pi}{T}\right)^2(0.080)^2$ $T = 0.401 \text{ s}$ <u>Method 2:</u> $\omega^2 = \frac{k}{m}$ $\left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$ $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.25}{61.4}} = 0.401 \text{ s}$ <u>Method 3:</u> At $y = 0.000 \text{ m}$, the extension of spring is maximum, $e_{\max} = 0.120 \text{ m}$. The mass is at its amplitude position in its SHM, so displacement is $x_o = 0.080 \text{ m}$. (See Fig. 4.1). $\uparrow +: F_{\text{net}} = ma \quad \Rightarrow \quad ke_{\max} - mg = ma_o$ $ke_{\max} - mg = m(\omega^2 x_o)$ $(61.4)(0.12) - (0.25)(9.81) = (0.25)(\omega^2 (0.08))$ $T = \frac{2\pi}{\omega} = 0.401 \text{ s}$	C1 A1